Rx Provenance & Quality Resolution Engine: UX & Backend System Specification

1. Critical Architectural Ground Truths (Read Before Coding)

Before implementing the resolution state machine, engineering must account for three structural traps in pharmaceutical regulatory data where naive logic fails:

1.1 The Intra-NDC Pill Imprint Fallacy

- **The Trap:** Assuming that if a known NDC maps to multiple physical manufacturing plants in its FDA Structured Product Labeling (SPL) XML, asking the user for the pill color and imprint will disambiguate which plant pressed the pill.
- **The Reality:** Under 21 CFR Part 206, a physical pill imprint maps to the active ingredient, strength, and the NDA/ANDA holder or Labeler (the 2-segment Product NDC). When a single Product NDC lists three approved Finished Dosage Form (FDF) plants in its SPL file, all three plants use identical tablet punches/dies and produce pills with the exact same imprint, color, and shape.
- **System Rule:**
    - Use **Pill Imprint + Color + Shape** ONLY to resolve an unknown `product_ndc` (e.g., when a retail amber bottle has no NDC, or when a corporate labeler holds multiple legacy ANDAs with different imprints for the same molecule and dose).
    - **NEVER** prompt a user for pill color/imprint to disambiguate multiple plants tied to an already-identified single `product_ndc`. If the user has an original factory bottle/carton, prompt for the printed **"Manufactured at / Made in [Location]"** fine print. If the user only has a retail pharmacy amber vial, deterministic single-plant resolution is impossible from public data; the backend must immediately execute **Worst-Case Facility Set Bounding**.

1.2 The Repackager / Relabeler Blind Spot (`<asEquivalentEntity>`)

- **The Trap:** Querying a valid 10- or 11-digit NDC and auditing only the establishments listed directly in that NDC's SPL XML.
- **The Reality:** Wholesalers and repackagers (e.g., Cardinal Health, Major Pharmaceuticals, Bryant Ranch Prepack) purchase bulk tablets from overseas manufacturers, repackage them into unit-dose blisters or smaller bottles, and assign their own Labeler NDC. Their SPL

XML often lists only a domestic packaging warehouse under the `REPACK` (C73606) or `RELABEL` (C73607) operation code.

- **System Rule:** If an SPL XML contains `REPACK` or `RELABEL` operations and lacks a direct `MANUFACTURE` (C43215) establishment, the backend parser MUST extract the `<asEquivalentEntity><definingMaterialKind><code>` node (the Source NDC) and recursively fetch the root manufacturer's SPL XML until it reaches the physical `MANUFACTURE` and `API MANUFACTURE` (C82401) establishments.

1.3 NDC Format Normalization (11-Digit HIPAA vs. 10-Digit FDA SPL)

- **The Trap:** Matching a scanned 11-digit pharmacy barcode or insurance claim NDC directly against openFDA or DailyMed SPL records.
- **The Reality:** Pharmacies, PBMs, and GS1 DataMatrix barcodes encode NDCs as an unhyphenated 11-digit string (`5-4-2` format) by padding a leading zero into one of the three segments. The FDA stores NDCs in 10-digit hyphenated formats (`4-4-2`, `5-3-2`, or `5-4-1`).
- **System Rule:** The backend normalizer must take any 11-digit input `L1L2L3L4L5-P1P2P3P4-K1K2`, strip the package segment (`K1K2`) to isolate the 9-digit candidate product code, and generate the three possible 10-digit hyphenated permutations by removing a leading zero from Segment 1 (if `L1 == 0`), Segment 2 (if `P1 == 0`), or Segment 3 (if `K1 == 0`), validating the resulting Labeler prefix against the official FDA NDC Labeler Directory before querying SPL.

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2. Master Resolution State Machine & Use Case Flows

The system ingests user input at whatever fidelity is available and drives toward a **Resolved Facility Set** (`{Plant_1, ..., Plant_k}`).

Use Case 1 (UC-1): Factory Stock Bottle, Sealed Carton, or Blister Pack

- **User Input:** 1D Barcode (UPC-A / GS1-128), 2D GS1 DataMatrix (DSCSA standard: GTIN-14 + Serial + Lot + Exp), or manual entry of the printed NDC on original manufacturer packaging.
- **Happy Path (Single Plant):**
    a. Scanner extracts GTIN-14 or 10/11-digit NDC.
    b. Backend normalizes to 2-segment `product_ndc`, resolves any `<asEquivalentEntity>` repackager links, and parses SPL `<performance>` blocks.

c. Backend filters establishments by NCI Thesaurus operation codes: `C43215` (`MANUFACTURE` - Finished Dosage Form) and `C82401` (`API MANUFACTURE`).
d. Exactly 1 active `C43215` facility is returned. Backend cross-references the facility's DUNS/FEI against FDA Inspection Classifications (NAI/VAI/OAI), Form 483s, Warning Letters, Import Alerts (specifically Import Alert 66-40 for cGMP failures), and Recall Enforcement Reports.
e. **UX Output:** Deterministic Plant Provenance Card (Single-Plant Verified).

- **Edge Case 1A: Multi-Plant Ambiguity on Factory Packaging ($k > 1$ FDF Plants in SPL)**
  - *Why it happens:* The NDA/ANDA holder registered multiple approved manufacturing sites under the same NDC to prevent shortages or shift volume globally.
  - *UX Disambiguation Prompt:* Because the user holds the **original factory container** (not a pharmacy amber vial), the physical label often prints the specific site of manufacture for that lot. Prompt the user:
    - *"Regulatory filings list [k] possible factories for this barcode. Since you have the original manufacturer package, snap a photo of the fine print near 'Manufactured by' or 'Made in', or select the printed location below: [Option A: City/Country 1] | [Option B: City/Country 2] | [Not Listed on Package]."* |---|If "Not Listed on Package", transition to **Worst-Case Facility Set Bounding (Section 3)**.
- **Edge Case 1B: Authorized Generics (AG) & NDA Shared Lines**
  - *What goes wrong:* An Authorized Generic is marketed under a generic labeler's NDC, and its SPL XML may only list the marketing distributor without enumerating the brand holder's plant.
  - *Backend Mitigation:* Join on `application_number` (e.g., `NDA021345`). Cross-reference the parent NDA's active SPL establishment roster when an Authorized Generic SPL omits `C43215` nodes.

Use Case 2 (UC-2): Retail Pharmacy Amber Vial — NDC Present on Label or Receipt

- **User Input:** Photo of a retail pharmacy vial (CVS, Walgreens, independent) or patient portal/insurance PDF receipt where an 11-digit or 10-digit NDC is printed (e.g., `NDC: 00093-1075-01`).
- **Backend Flow:**
  a. OCR extracts the NDC string and also extracts the printed Drug Name + Strength as a sanity check.

b. **Cross-Validation Check:** Backend verifies that the extracted NDC's active ingredient and strength in openFDA match the OCR-extracted drug name and strength on the label.
    - *Failure Mode:* Pharmacy label templates contain several numbers (Store #, Rx #, Date, Internal Bin #, Zip Code) that OCR regex can misclassify as an NDC, or a single compromised digit changes the drug entirely. If NDC lookup != Printed Drug Name, discard the OCR NDC and drop into **UC-3**.
c. Backend queries SPL establishments for the verified `product_ndc` (recursing through `<asEquivalentEntity>` if the pharmacy dispensed from a repackager bulk bottle).

- **Edge Case 2A: Multi-Plant Ambiguity on an Amber Vial ($k > 1$ FDF Plants in SPL)**
    - *Critical Distinction from UC-1A:* The user has a **generic orange pharmacy vial**, NOT the factory stock bottle.
        i. The orange vial does NOT print the factory address or country of origin.
        ii. Asking for the pill color/imprint is **useless** because all k plants under this `product_ndc` stamp the exact same imprint.
    - *UX Output (No Dead-End Prompts):* Do NOT prompt the user for pill imprint or bottle fine print. Immediately render the **Multi-Plant Risk Bounded View**:
        - Display all k approved plants tied to this NDC.
        - Compute the safety status using **Worst-Case Bounding** (if Plant A is clean/NAI and Plant B has an active OAI/Warning Letter, score the medication at Plant B's risk level).
        - If the drug is a **Narrow Therapeutic Index (NTI)** medication (e.g., levothyroxine, warfarin, tacrolimus, phenytoin, lithium), display an explicit **Multi-Plant Variance Warning**: *"Even though your NDC is known, this labeler rotates production across [k] different factories under the same barcode. For narrow-margin drugs, silent plant switches can alter bioavailability. Consider requesting a single-plant manufacturer (View Verified Single-Plant Alternatives)."*

Use Case 3 (UC-3): Retail Pharmacy Amber Vial — No NDC, Manufacturer Text Present

- **User Input:** Photo of a retail amber vial with NO NDC, containing Drug Name, Strength, Dosage Form, and a truncated pharmacy manufacturer string (e.g., `LEVOTHYROXINE 75 MCG TAB`, `MFR: AUROBINDO` or `MFR: MYL` or `DIST BY: ASCEND`).
- **Backend Flow & Failure Modes:**

a. **OCR & Entity Normalization:** Extract molecule, strength, form, and raw manufacturer string.
b. **The Pharmacy Mnemonic Problem:** Retail pharmacy dispensing systems (CVS RxConnect, Walgreens Intercom Plus, PioneerRx, Epic Willow) use truncated 3-to-8 character strings (`AUR`, `DRL`, `LUP`, `ZDU`, `CBER`, `TEV`, `ACT`). A naive exact-string match against FDA's `openfda.manufacturer_name` will fail.
    - *Backend Solution:* Maintain a curated **Pharmacy Mnemonic-to-FDA Labeler Code Mapping Table** combined with trigram similarity matching filtered strictly to labelers holding active `product_ndc` registrations for that specific RxNorm Concept Unique Identifier (`RxCUI`).
c. **Post-Match Cardinality Check (How Many `product_ndc`s Match?):**
    - **Case 3A (1 Labeler → 1 Active `product_ndc` for that RxCUI):** Lock onto that `product_ndc` automatically and evaluate its SPL facility roster (proceeding to deterministic output or Worst-Case Bounding if multi-plant).
    - **Case 3B (1 Labeler → Multiple Active `product_ndc`s for that RxCUI):**
        - *Why it happens:* Corporate M&A (e.g., Teva owns legacy Teva, Actavis, and Watson ANDAs for the same molecule/dose; Sun Pharma owns Ranbaxy and Taro ANDAs) or a company markets both an AB1 and AB2 rated version.
        - *Can Pill Imprint Disambiguate Case 3B?* **YES.** Because these are distinct `product_ndc` / ANDA filings under the same corporate parent, they almost always have **distinct physical pill imprints**.
        - *UX Disambiguation Prompt: "We identified [Manufacturer Name], which holds [N] different FDA registrations for [Drug + Dose]. Which imprint is stamped on your pill? [Option 1: Imprint A (Shape/Color)] | [Option 2: Imprint B (Shape/Color)]."*
        - Once the user taps their imprint, lock the exact `product_ndc` and audit its SPL plant roster.

Use Case 4 (UC-4): "Black Box" Retail Amber Vial — Zero NDC, Zero Manufacturer Info

- **User Input:** Amber vial label contains ONLY the generic drug name and strength (e.g., `WARFARIN SODIUM 5 MG TABLET`), or the user only has loose pills in a pill organizer.
- **UX & Backend Flow:**

a. **Context Lock:** Lock the `RxCUI` (Ingredient + Strength + Dosage Form) from the label OCR or text search.
b. **Candidate Set Pre-Fetch:** Query the backend database (joined across DailyMed SPL indexing + NIH RxNav/RxTerms + openFDA) for all active `product_ndc` records matching that `RxCUI`.
c. **Dynamic Faceted Pill Narrowing (Do NOT use open-ended text boxes first):**
    - *Why open-ended CV/text fails:* White-on-white debossed (un-inked) stamps on 6mm tablets produce high OCR failure rates due to specular glare and shadow inversion. Users also confuse `0`/`O`, `1`/`I`/`l`, and score lines.
    - *UX Solution (Constrained Selection):* Because the backend already knows the drug is `Levothyroxine 75 mcg`, there are only ~8–15 valid legal pill imprints in the entire US market for that dose.
    - **Step 4.1 (Color Filter):** Show only the colors that actually exist for `Levothyroxine 75 mcg` (e.g., Violet, White, Purple). User taps **Violet**.
    - **Step 4.2 (Candidate Imprint Cards):** Filter the 12 candidates down to the 3–4 matching that color, and render high-contrast visual diagrams showing **Side A / Side B imprints and score lines** (e.g., `[ M | L 15 ]`, `[ L75 | blank ]`, `[ E 5 | blank ]`) plus an option to type custom characters if not listed.
d. **Backend Resolution & Edge Cases:**
    - **Edge Case 4A: Repackager Imprint Multiplicity:** Searching `Imprint = M; L 15` in openFDA returns 15 different `product_ndc`s because 14 repackagers sell Mylan's pill under their own NDC while keeping Mylan's physical imprint (`M | L 15`).
        - *Backend Mitigation:* Deduplicate all matching NDCs by traversing `<asEquivalentEntity>` to the root NDA/ANDA `application_number` and original manufacturer `product_ndc`. All 14 repackagers collapse into 1 root manufacturing profile.
    - **Edge Case 4B: Authorized Generic Imprint Collision:** A brand drug (e.g., Synthroid) and its Authorized Generic share the exact same physical pill imprint, color, and factory line, differing only by bottle label.
        - *Backend Mitigation:* Since both share the identical NDA and physical manufacturing plant, collapsing them by `application_number` + `SPLIMPRINT` yields the exact same physical plant audit.

Use Case 5 (UC-5): Unresolvable / Opaque Supply Chain / Compounded Drugs

- **Scenarios where resolution hits a hard wall:**
  a. **Missing SPL Establishment Data:** ~5–10% of active SPL XML files on DailyMed either omit the `<performance>` establishment block entirely or list only a corporate headquarters DUNS (`LABEL` / `ANALYSIS`) with zero `MANUFACTURE` (C43215) facility listed.
  b. **503A / 503B Compounded Medications:** Custom compounded capsules/creams or tele-health compounded GLP-1s/hormones do not go through the ANDA/NDA SPL establishment pipeline.
  c. **Over-Encapsulated or Unlisted Imprints:** Newly approved generics whose SPL XML hasn't synced to openFDA/DailyMed yet (up to 14–30 day lag), or foreign/unapproved pills.
- **System Rule & UX Output:**
  - Never fail silently and never show a "Green / Clean" status when facility records are empty (`establishments.length == 0` for `C43215`). A plant with zero inspection records is an **Unverified / Opaque Risk**, not a safe plant.
  - Render an **Opaque Provenance Alert**: *"This labeler's FDA regulatory filing omits the physical manufacturing plant identifier (FEI/DUNS). Under our safety model, undisclosed manufacturing facilities cannot be verified as compliant."*

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3. Backend Data Pipeline & Failure Mode Reference Table

Subsystem	Data Source / Field	What Goes Wrong in Production	Required Engineering Mitigation
SPL XML Facility Parsing	DailyMed SPL XML `<assignedEntity>` & `<performance><actDefinition><code code="..."/>`	SPL lists testing labs (`C25391`), packagers (`C84731`), API makers (`C82401`), and FDF makers (`C43215`) in the same list.	Strictly partition establishments by NCI Thesaurus code: `C43215` = Finished Dose Plant; `C82401` = API Chemical Plant; `C73606`/`C73607` = Repackager/Relabeler.
Dormant Facility Bloat	DailyMed SPL XML	Manufacturers leave backup or decommissioned contract plants in their SPL XML for years so they don't lose regulatory approval to use them.	FDA does not publish batch-level plant volume splits. Treat all listed `C43215` plants in an active SPL as active candidates;

Subsystem	Data Source / Field	What Goes Wrong in Production	Required Engineering Mitigation
			surface "Multi-Plant Network (k sites)" as an explicit supply-chain fragility metric.
DUNS to FEI Mapping	SPL uses 9-digit DUNS (`1.3.6.1.4.1.519.1`); FDA Inspections Database (OAI/VAI/483) uses 7-to-10 digit FDA **FEI**.	Direct join between SPL XML and FDA Data Dashboard fails because SPL rarely includes the FEI directly.	Build an automated daily ETL crosswalk joining: (1) FDA CDER Drug Establishments Current Registration Site (`DECRS` file, which contains both DUNS and FEI), (2) GDUFA Facility Lists, and (3) normalized address/geo-coordinate fallback matching.
Inspection Lag & Import Alerts	FDA Data Dashboard API (`inspections_classifications`, `compliance_actions`)	A plant in India or China receives an FDA Import Alert (`66-40`) or Form 483 months before an Official Action Indicated (OAI) classification is finalized in the API.	Ingest three distinct feeds: (1) FDA Inspection Classification API, (2) FDA Import Alert 66-40 parser (HTML/XML scrape of DWPE list, accounting for "carve-out" exceptions for shortage drugs), and (3) openFDA Recall Enforcement API.
NTI Classification	RxNorm / FDA Orange Book Therapeutic Equivalence (`TE`) Codes	Standard generics (`AB` rated) allow an 80%–125% bioequivalence confidence interval, which is dangerous to switch across plants for Narrow Therapeutic Index drugs.	Flag any `RxCUI` belonging to the NTI watchlist (Levothyroxine, Warfarin, Liothyronine, Tacrolimus, Cyclosporine, Phenytoin, Carbamazepine, Lithium, Digoxin, Theophylline) or multi-rated generics (`AB1`, `AB2`, `AB3`, `BX`). Enforce strict Single-Plant preference rules on these molecules.

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4. Developer Decision Tree (Pseudocode Specification)

```
FUNCTION ResolveMedicationProvenance(userInput):


 // STEP 1: IDENTIFY PRODUCT_NDC
 IF userInput.hasBarcodeOrNDC:
  candidateNDCs = NormalizeTo10DigitProductNDCs(userInput.rawNDC)
  IF userInput.hasLabelText:
   candidateNDCs = FilterByMatchingIngredientAndStrength(candidateNDCs, userInput.labelText)
  IF candidateNDCs.isEmpty():
   GOTO Step2_LabelTextResolution
  productNDC = candidateNDCs.first()
 ELSE IF userInput.hasLabelText:
  Step2_LabelTextResolution:
  rxCUI = ResolveRxCUI(userInput.drugName, userInput.strength, userInput.dosageForm)
  IF userInput.hasMfrString:
   labelerCodes = MatchPharmacyMnemonicToLabeler(userInput.mfrString, rxCUI)
   matchingProductNDCs = GetActiveProductNDCs(rxCUI, labelerCodes)


   IF matchingProductNDCs.count == 1:
    productNDC = matchingProductNDCs[0]
```

```
    ELSE IF matchingProductNDCs.count > 1:
      // Same corporate parent has multiple ANDAs with different imprints
      distinctImprints = GetImprintProfiles(matchingProductNDCs)
      selectedImprint = PROMPT_USER_SELECT_IMPRINT(distinctImprints)
      productNDC = MapImprintToProductNDC(matchingProductNDCs, selectedImprint)
    ELSE:
      GOTO Step3_PhysicalPillResolution
  ELSE:
    GOTO Step3_PhysicalPillResolution
ELSE:
  Step3_PhysicalPillResolution:
  rxCUI = ResolveRxCUI(userInput.drugName, userInput.strength, userInput.dosageForm)
  validImprintProfiles = GetAllMarketImprintsForRxCUI(rxCUI)
  selectedProfile = PROMPT_USER_FACETED_PILL_PICKER(
    colors = validImprintProfiles.distinctColors(),
    shapes = validImprintProfiles.distinctShapes(),
    imprintCards = validImprintProfiles
  )
  // Collapse repackagers sharing the same physical pill imprint via <asEquivalentEntity>
  productNDC = ResolveRootSourceProductNDC(selectedProfile)
```

```
  // STEP 2: TRAVERSE REPACKAGER LINKS TO ROOT SPL
  splDocument = FetchSPLByProductNDC(productNDC)
  WHILE splDocument.has only REPACK/RELABEL and has <asEquivalentEntity>:
    sourceProductNDC = splDocument.getEquivalentSourceNDC()
    splDocument = FetchSPLByProductNDC(sourceProductNDC)
  // STEP 3: EXTRACT & PARTITION FACILITIES
  fdfPlants = splDocument.getEstablishments(operation = "C43215") // MANUFACTURE
  apiPlants = splDocument.getEstablishments(operation = "C82401") // API MANUFACTURE
  IF fdfPlants.isEmpty() AND splDocument.isAuthorizedGeneric():
    parentNDASPL = FetchSPLByApplicationNumber(splDocument.applicationNumber)
    fdfPlants = parentNDASPL.getEstablishments(operation = "C43215")
    apiPlants = parentNDASPL.getEstablishments(operation = "C82401")
  // STEP 4: HANDLE PLANT CARDINALITY & AMBIGUITY
  IF fdfPlants.count == 0:
    RETURN RenderOpaqueProvenanceWarning(productNDC,
reason="MISSING_SPL_C43215_NODE")
  ELSE IF fdfPlants.count == 1:
    plantAudit = AuditFacilityByDUNSandFEI(fdfPlants[0], apiPlants)
    RETURN RenderDeterministicReport(productNDC, fdfPlants[0], plantAudit)
  ELSE IF fdfPlants.count > 1:
```

```
  // CRITICAL RULE: Do NOT ask for pill imprint here (all fdfPlants share the same imprint).
  IF userInput.containerType == "ORIGINAL_FACTORY_PACKAGE":
    printedSite = PROMPT_USER_FACTORY_FINE_PRINT(fdfPlants.locations())
    IF printedSite != "NOT_LISTED":
      lockedPlant = fdfPlants.filterByLocation(printedSite)
      RETURN RenderDeterministicReport(productNDC, lockedPlant,
AuditFacilityByDUNSandFEI(lockedPlant, apiPlants))

  // Retail amber vial OR factory package without site text → Worst-Case Set Bounding
  multiPlantAudit = ComputeWorstCaseBoundedAudit(fdfPlants, apiPlants)
  isNTI = CheckIfNarrowTherapeuticIndex(rxCUI)
  RETURN RenderMultiPlantBoundedReport(
    productNDC = productNDC,
    candidatePlants = fdfPlants,
    worstCaseScore = multiPlantAudit,
    showNTIVarianceWarning = isNTI,
    singlePlantAlternatives = GetVerifiedSinglePlantNDCs(rxCUI)
  )
```